

Agilex 5 Camera 4Kp60 HDR

One-Time Bring-up, Offline Customer Demo, and Build Guide

Field	Value
Target design	4Kp60 Multi-Sensor HDR Camera Solution System Example Design
Target hardware	Agilex 5 FPGA E-Series 065B Modular Development Kit
Default documentation release	26.1
Intended audience	Engineer bringing up the kit for the first time or demonstrating it to a customer
Primary use case	Run the pre-built image; build from source only when modification is required

This is an Altera example design. It is a development demonstration, not a production-qualified camera product.

1. What this design does

The design accepts one or two Framos FSM:GO IMX678 camera inputs over MIPI CSI-2, processes the images in the FPGA ISP pipeline, and drives 4Kp60 video over DisplayPort. Linux runs on the HPS and starts the application, camera control, and browser GUI.

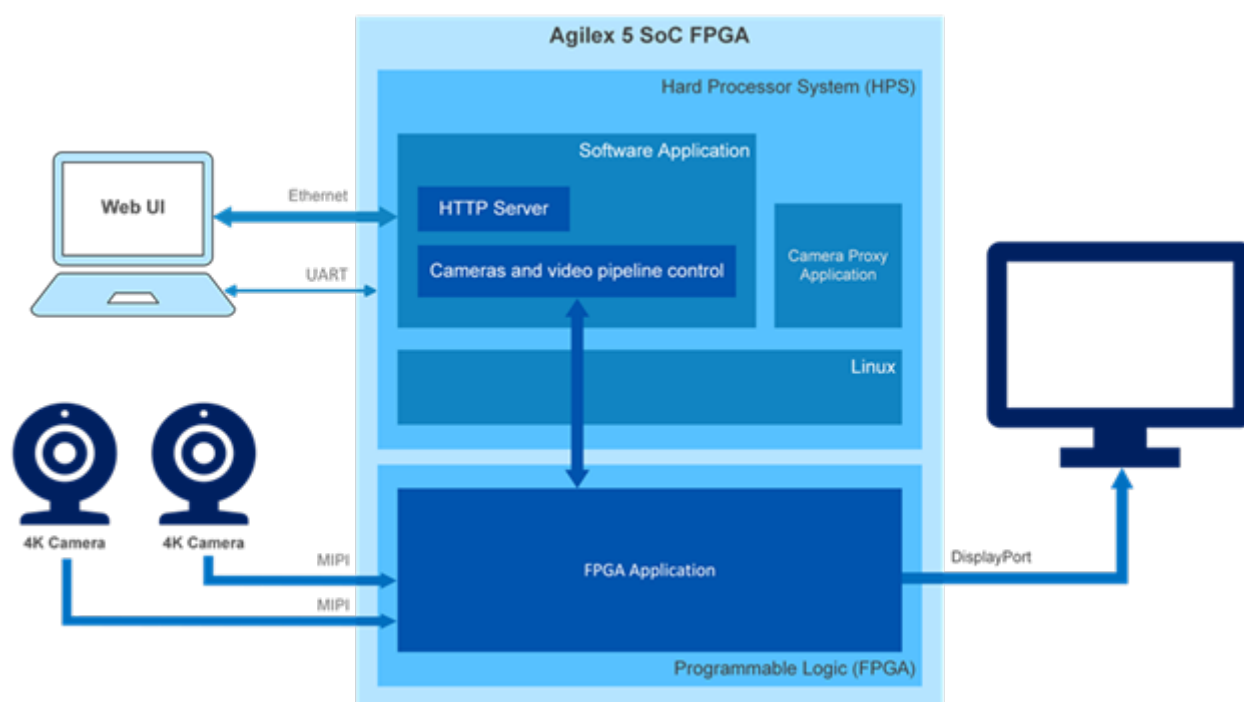


Figure 1. Official Altera high-level block diagram. Source: [26.1 camera 4K design guide](#).

Expected final result

1. The monitor connected to **Carrier DisplayPort TX** shows the ISP-processed camera output.
 2. The HPS Linux console is available through the SOM UART USB connection.
 3. A browser on the host PC opens `http://<board-ip>/` and displays the camera GUI.
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2. Official links and downloads

Run the pre-built design

Purpose	Official link
26.1 camera design overview	https://altera-fpga.github.io/rel-26.1/embedded-designs/agilex-5/e-series/modular/camera/camera_4k/camera_4k/
Detailed 26.1 bring-up guide	https://github.com/altera-fpga/agilex5-ed-camera/blob/rel/26.1/docs/es/camera/camera_4k/camera_4k.md
26.1 pre-built release assets	https://github.com/altera-fpga/agilex5-ed-camera/releases/tag/rel-26.1-isp_hdr-MDK_RevB_GrpB
26.1 QSPI image (<code>top.core.jic</code>)	https://github.com/altera-fpga/agilex5-ed-camera/releases/download/rel-26.1-isp_hdr-MDK_RevB_GrpB/top.core.jic
26.1 microSD image (<code>.wic.gz</code>)	https://github.com/altera-fpga/agilex5-ed-camera/releases/download/rel-26.1-isp_hdr-MDK_RevB_GrpB/hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic.gz
25.1 pre-built release assets (alternate test release)	https://github.com/altera-fpga/agilex5-ed-camera/releases/tag/rel-25.1
Quartus Prime Pro 26.1 download	https://www.altera.com/downloads/fpga-development-tools/quartus-prime-pro-edition-design-software-version-26-1-linux

Build or modify the design

Purpose	Official link
Camera design source repository, rel/26.1 branch	https://github.com/altera-fpga/agilex5-ed-camera/tree/rel/26.1
Hardware build / MDT guide	https://github.com/altera-fpga/agilex5-ed-camera/blob/rel/26.1/AGX_5E_Altera_Modular_Dk_ISP_designs/HDR_CAMERA.md
Modular Design Toolkit, rel/26.1	https://github.com/altera-fpga/modular-design-toolkit/tree/rel/26.1
Software / Yocto build guide	https://github.com/altera-fpga/agilex5-ed-camera/blob/rel/26.1/sw/README.md
Git LFS installation	https://git-lfs.com/
Yocto Project requirements	https://docs.yoctoproject.org/ref-manual/system-requirements.html
KAS installation and dependencies	https://kas.readthedocs.io/en/latest/userguide.html#dependencies-installation
VVP IP Suite	https://www.altera.com/products/ip/po-3150/video-and-vision-processing-suite
MIPI D-PHY / CSI-2 IP	https://www.altera.com/products/ip/po-3062/mipi-d-phy-ip

3. Select the correct release pair

The FPGA/QSPI and Linux/microSD images must come from the **same release**.

Do	Do not
Use 26.1 <code>top.core.jic</code> with the matching 26.1 <code>.wic.gz</code> image	Mix a 26.1 JIC with a 25.1 microSD image
Store release files in separate host folders	Rename files and lose release traceability
Record the release used on the kit	Assume a new microSD image works with existing QSPI contents

For the default 26.1 pre-built flow, the two required files are:

```
top.core.jic
hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic.gz
```

4. Hardware and software checklist

Hardware

Item	Required
Agilex 5 E-Series 065B Modular Development Kit	Yes
Framos FSM:GO IMX678 camera module(s) and compatible lens	One or two
Framos MIPI cable for each camera	Yes
microSD card, 8 GB minimum	Yes
DisplayPort cable and 4K-capable monitor	Yes
Micro-USB cable for Carrier J35 (JTAG programming)	Yes for one-time QSPI programming
Micro-USB cable for SOM J2 (HPS serial console)	Yes
Ethernet cable for SOM J6	Yes
Host PC running Linux or Windows	Yes

Host software

Tool	Purpose
Quartus Prime Pro Programmer and Tools	Program the QSPI JIC
minicom, GtTerm, PuTTY, or Tera Term	HPS serial console
Browser (Chrome, Edge, Firefox)	Camera GUI
dd (Linux) or an SD-card imaging tool	Write the microSD image

Work with the development kit, microSD card, and camera modules at an ESD-safe workstation.

5. One-time bring-up: program the kit from scratch

Use this section when the kit has no prepared microSD card and/or no matching QSPI image.

5.1 Prepare a local release folder

Download both files from the release page before starting. Keep them locally for offline bring-up later.

```
mkdir -p ~/agilex5-camera-26.1
cd ~/agilex5-camera-26.1

# Download from the official links in Section 2, then confirm both files exist.
ls -lh top.core.jic \
    hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic.gz
```

Optional: create a local checksum record after download:

```
sha256sum top.core.jic \
    hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic.gz \
    > release-file-sha256.txt
```

5.2 Create the bootable microSD card

The release image is a gzip-compressed disk image. Extract it:

```
cd ~/agilex5-camera-26.1
gunzip -k hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic.gz
```

Insert the microSD card reader and identify the removable device:

```
lsblk
```

Use the **whole removable disk** (for example `/dev/sda`), not a partition such as `/dev/sda1`, and never the host system disk (often `nvme0n1`).

```
# Example only: replace /dev/sdX with the microSD device confirmed above.
sudo umount /dev/sdX1
sudo dd if=hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic \
    of=/dev/sdX bs=1M status=progress conv=fsync
sync
sudo eject /dev/sdX
```

Power off the kit and insert the microSD card into the **SOM microSD slot**.

5.3 Program the QSPI flash

This is normally required only once for a chosen release.

1. Power the kit **off**.
2. Set the SOM **MSEL/S4 switch to OFF-OFF** (JTAG configuration mode).
3. Connect Carrier **J35** Micro-USB to the host PC.
4. Power the kit **on**.
5. Confirm JTAG is detected:

```
jtagconfig
```

1. Program the matching JIC:

```
cd ~/agilex5-camera-26.1
quartus_pgm -c 1 -m jtag -o "pvi;top.core.jic"
```

If `jtagconfig` lists a cable index other than `1`, replace `-c 1` with the listed index. Wait for the Programmer success message.

1. Power the kit **off**.
2. Set the SOM **MSEL/S4 switch to ON-ON** (AS×4 / QSPI boot mode).

5.4 Connect the final system

With the kit powered off:

1. Connect one or two cameras to the carrier MIPI connector(s), aligning **pin 1 to pin 1**.
2. Connect Carrier **DisplayPort TX** to the monitor.
3. Connect SOM **J2** to the host PC for the HPS serial console.
4. Connect SOM **J6** Ethernet to the same network as the host PC.
5. Power on the kit and wait about two minutes for Linux and the application to start.

6. Customer-facing offline bring-up

Use this section at a customer site where Internet access may be unavailable.

6.1 What to prepare before travelling

Item	Offline requirement
Host laptop	Quartus Programmer, serial terminal, and browser already installed
Release folder	Matching <code>top.core.jic</code> and <code>.wic</code> image retained locally
Prepared microSD	Written and verified with the matching <code>.wic</code> image
Cables	J35 USB, J2 USB, J6 Ethernet, MIPI, and native DisplayPort
Monitor	Tested to accept 4Kp60 DisplayPort input
Network	Direct Ethernet cable to host or a local DHCP router/switch; Internet is not required

6.2 First-time customer-site deployment

1. Check the microSD card is inserted in the SOM slot.
2. Power off the board and set **MSEL/S4 = OFF-OFF**.
3. Connect **J35** to the host, power on, and program the locally stored matching JIC:

```
cd ~/agilex5-camera-26.1
jtagconfig
quartus_pgm -c 1 -m jtag -o "pvi;top.core.jic"
```

1. When programming succeeds, power off and set **MSEL/S4 = ON-ON**.
2. Connect the camera(s), DisplayPort TX monitor, J2 USB serial, and J6 Ethernet.
3. Power on and continue with Section 6.4.

6.3 Repeat demo on a pre-provisioned kit

When matching QSPI and microSD are already installed, do **not** reprogram QSPI:

1. Confirm **MSEL/S4 = ON-ON**.
2. Insert the prepared microSD card.
3. Connect MIPI camera(s), DP TX monitor, J2 USB, and J6 Ethernet.
4. Power on and wait about two minutes.
5. Continue with Section 6.4.

6.4 Open the HPS terminal

The lab-confirmed command below opens the serial console in a separate terminal window:

```
gnome-terminal --title="Agilex-HPS-ttyUSB3" -- bash -c \  
"sudo minicom -D /dev/ttyUSB3 -b 115200; exec bash"
```

Serial settings:

```
115200 baud, 8 data bits, 1 stop bit, no parity,  
hardware flow control disabled, software flow control disabled
```

Press Enter. At the login prompt, enter:

```
root
```

`/dev/ttyUSB3` is the proven lab mapping, but Linux assigns ttyUSB numbers during USB enumeration. If it is blank or unavailable, run `ls -l /dev/ttyUSB*` and identify the serial ports added by **SOM J2**. The official guidance is to use the HPS UART port from that J2 group. If J35 is also connected, its serial ports may enumerate first.

6.5 Network without Internet access

The browser GUI works over local Ethernet; Internet access is not required after the design assets are prepared.

Option A: local router or switch with DHCP

Connect both the host and board J6 to the same local DHCP-capable router/switch. On the board:

```
ip -4 addr show eth0
```

Record the `eth0` address.

Option B: direct board J6 to Ubuntu host

On Ubuntu, configure the wired Ethernet connection as **Shared to other computers**:

Settings → Network → Wired → IPv4 → Shared to other computers

Confirm the host obtains an address similar to:

```
ip -4 addr show enp3s0
# Expected example: 10.42.0.1/24
```

On the board, obtain the address:

```
ip -4 addr show eth0
```

If no IPv4 address appears, request a DHCP lease:

```
udhcpc -i eth0
ip -4 addr show eth0
```

Keep the default Ethernet MTU of **1500**. The original board investigated in this lab had a physical size-dependent Ethernet fault; lowering MTU only hid the fault and made the GUI unreliable. A healthy board passes the test in Section 7 at MTU 1500.

6.6 Open the camera GUI

On the host:

```
BOARD_IP=<board-eth0-ip>
google-chrome-stable http://$BOARD_IP/
```

The browser should show the Altera splash screen, then the camera GUI.

7. Customer-demo acceptance test

Run this before a demonstration. It verifies the complete board-to-host Ethernet path at the standard MTU.

```
BOARD_IP=<board-eth0-ip>

ping -c 2 $BOARD_IP
ping -c 2 -M do -s 1472 $BOARD_IP
curl -sS --max-time 10 http://$BOARD_IP/ | wc -c
```

Pass criteria:

Check	Expected result
Small ping	Two replies
Full-size ping	Two replies at payload size 1472
HTTP download	Nonzero HTML byte count
Browser	Altera splash then responsive GUI
Monitor	Camera or test-pattern video on DisplayPort TX

If the full-size ping fails or browser receives no HTML:

1. Keep MTU at 1500.
2. Try a known-good Ethernet cable.
3. Test against a second host PC.
4. If the failure follows the board, remove the board from demo use and open a board support case.

8. Camera and DisplayPort operation

Select the camera

In the browser GUI:

```
Input Config → Input Source → Camera
```

When using one camera on MIPI1 and video is missing, use the documented workaround:

```
Camera → Input TPG → Camera
```

Verify camera detection from HPS

```
grep -iE 'Found camera|Active input|GMSL' /tmp/vvp-isp-demo.log | head -n 30
```

Healthy log examples include:

```
Found camera: FSM-IMX678. Enumerated as [Cam 0]
Active input: Camera 0
[Camera 0] new video standard: 3840x2160p @60 RGB 12 bit
```

DisplayPort no-signal troubleshooting

1. Confirm cable is in carrier **DisplayPort TX**, not another connector.
2. Prefer a native DisplayPort cable. A DP-to-HDMI adapter must support 4Kp60.
3. In the GUI, select **Input TPG**. If a test pattern appears, DisplayPort works; switch back to **Camera**.
4. Reseat the DP cable, select the correct monitor input, and power-cycle the monitor if needed.

9. Build from source

Use this section only when hardware or software changes are required. For a customer demo, use pre-built assets from Section 2.

9.1 Choose a hardware build flow

Flow	License/use case	Result
SOF MDT flow	OpenCore Plus evaluation, time-limited or JTAG-tethered evaluation	.sof
RBF MDT flow	Full IP licenses, turnkey microSD/QSPI deployment	.rbf + .jic
Pre-generated Quartus project	Explore implementation and reports	Quartus project

The RBF/QSPI flow requires full licenses for VVP Suite, Tone Mapping Operator, Warp, and 3D LUT IP. MIPI D-PHY, MIPI CSI-2, and Nios V licenses are also required as documented by Altera.

9.2 Clone the source

Install Git LFS before cloning. The repository includes a submodule.

```
git clone -b rel/26.1 --recurse-submodules \
  https://github.com/altera-fpga/agilex5-ed-camera.git \
  agilex5-ed-camera
```

9.3 Create the hardware project with MDT

SOF / OpenCore Plus evaluation:

```
cd agilex5-ed-camera
quartus_sh -t ./modular-design-toolkit/scripts/create/create_shell.tcl \
  -proj_path <project-path> \
  -proj_name agilex5_modkit_vvpisp \
  -xml_path ./AGX_5E_Altera_Modular_Dk_ISP_designs/AGX_5E_Modular_Devkit_ISP_FF_RD.xml
```

RBF / fully licensed flow:

```
cd agilex5-ed-camera
quartus_sh -t ./modular-design-toolkit/scripts/create/create_shell.tcl \
  -proj_path <project-path> \
  -proj_name agilex5_modkit_vvpisp \
  -xml_path ./AGX_5E_Altera_Modular_Dk_ISP_designs/AGX_5E_Modular_Devkit_ISP_RD.xml
```

9.4 Build the hardware project

SOF / evaluation:

```
cd <project-path>/scripts
quartus_sh -t build_shell.tcl -update_ocs -full_compile -ff_post_agx5e
```

RBF / fully licensed:

```
cd <project-path>/scripts
quartus_sh -t build_shell.tcl -update_ocs -full_compile -hps_post_agx5e
```

Expected outputs:

Flow	Output
SOF	fsbl_agilex5_modkit_vvpisp_time_limited.sof
RBF	agilex5_modkit_vvpisp.hps_first.hps.jic and agilex5_modkit_vvpisp.hps_first.core.rbf

9.5 Build the Linux microSD image

The official software flow needs a Linux host (or Docker) with at least **70 GB** free storage and **32 GB** RAM.

For the fully licensed HPS-first design:

```
cd agilex5-ed-camera/sw
KAS_MACHINE=agilex5_mk_a5e065bb32aes1 kas build kas/agilex_camera.yml
```

For OpenCore Plus FPGA-first evaluation:

```
cd agilex5-ed-camera/sw
KAS_MACHINE=agilex5_mk_a5e065bb32aes1 kas build kas/agilex_camera_ff.yml
```

The generated `.wic.gz` appears under:

```
build/tmp/deploy/images/agilex5_mk_a5e065bb32aes1/
```

If you build custom hardware, use a matching custom RBF/JIC and software image. Changing the hardware can cause the stock application image to fail.

10. Troubleshooting table

Symptom	Likely cause	Action
No <code>/dev/ttyUSB*</code>	Board off or USB cable missing	Power on; connect SOM J2
Serial terminal blank	Incorrect host serial device	Use the J2 UART group; try the known <code>ttyUSB3</code> mapping first
Only Nios banner visible	Connected to a non-HPS serial interface	Select the J2 HPS UART port
<code>eth0</code> has no IPv4	No DHCP path	Enable host Ethernet sharing or connect a DHCP router/switch; run <code>udhcpc -i eth0</code>
Small ping works but full-size ping fails	Ethernet path/board fault	Keep MTU 1500; follow Section 7 board triage
Browser blank, local server works	Ethernet path not carrying full response	Run Section 7; do not lower MTU to hide packet loss
Camera detected but monitor blank	DisplayPort issue	Use Input TPG, check DP TX, cable, monitor input
Single MIPI1 camera has no video	Documented design issue	Toggle Camera → Input TPG → Camera
Wrong release booted	Mismatched JIC and microSD	Reflash matching release pair

11. One-page customer-demo checklist

BEFORE TRAVEL

- [] Matching JIC and microSD image saved locally
- [] microSD already written
- [] Quartus Programmer + minicom installed
- [] J35, J2, J6, MIPI, and DP cables packed
- [] Monitor accepts 4Kp60 DP

FIRST-TIME KIT PREPARATION

- [] Power off; MSEL/S4 = OFF-OFF
- [] J35 connected; power on
- [] Program matching `top.core.jic`
- [] Power off; MSEL/S4 = ON-ON
- [] Insert matching microSD

CUSTOMER DEMO

- [] Connect camera(s), DP TX monitor, J2 serial, J6 Ethernet
- [] Power on; wait about 2 minutes
- [] Open HPS terminal:


```
gnome-terminal --title="Agilex-HPS-ttyUSB3" -- bash -c \
  "sudo minicom -D /dev/ttyUSB3 -b 115200; exec bash"
```
- [] Login: root
- [] Run: `ip -4 addr show eth0`
- [] Browser: `http://<board-ip>/`
- [] GUI: Input Config → Input Source → Camera
- [] Confirm 1472-byte ping before demo

12. Revision history

Version	Change
1.0	Initial standalone one-time bring-up, offline demo, and build guide